

Aditya Kulkarni

📍 Raleigh, NC | 📞 (281) 876-7066 | ✉️ adityakulkarnius@gmail.com
🌐 linkedin.com/in/aditya-kulkarni-355b81217 | 🐙 github.com/Aditya-k24

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To the Integrated Biosciences Engineering Team,

Drug discovery has always been constrained by the pace at which scientists can design, run, and interpret experiments. Integrated Biosciences is attacking that constraint directly—building AI-native infrastructure that puts intelligent automation in the hands of researchers. I am an engineer who has spent the last year shipping production agentic AI systems, and I have built on biological research data. That combination is not common.

- **MCP Tooling & Agentic Orchestration:** At **Isomer AI**, I deployed 10+ production AI features built on the Model Context Protocol (MCP) and Temporal.io agentic workflows—multi-model pipelines spanning Claude, GPT-4o, and Gemini across 923 commits and 7 production deployments. I use Claude Code daily as my primary development environment; I did not need to learn it for this role.
- **Biological Research Data at Scale:** As a **Graduate Research Assistant at NC State**, I built a cloud-native platform on GCP serving 500GB+ of genomic datasets to 100+ researchers, and improved ML model accuracy by 12% through hyperparameter tuning and feature engineering on biological sensor data. Most software engineers have never touched biological research data—I built infrastructure specifically to serve scientists who work with it.
- **Full-Stack TypeScript / React / PostgreSQL:** My production stack at Isomer AI is TypeScript, Node.js, PostgreSQL, Redis, and Docker—the exact combination this role requires. I own schema design, query optimization, CI/CD, and incident response across enterprise client environments.
- **Production AI Infrastructure:** I built **CortexQ**, a Kubernetes-native LLM inference platform on GCP with KEDA autoscaling, multi-provider model serving, and full OpenTelemetry observability—load tested to 500 concurrent users and validated with 88 integration tests. Building reliable AI infrastructure at scale is work I do because I find it genuinely interesting.

Integrated Biosciences is compressing the timeline between a biological insight and a medicine that reaches patients. I want to build the software that makes that faster. I would be glad to discuss how my experience maps to your team's roadmap.

Sincerely,

Aditya Kulkarni